

SMART CONTRACT-DRIVEN ENERGY TRADING MECHANISM USING BLOCKCHAIN

Richa¹, Dr. Ajay Kumar²

¹ Student, Department of Computer Science & Engineering, Chitkara University Institute of engineering & Technology, Punjab, India.

Richa2150.be22@chitkara.edu.in

² Associate Professor, Department of Computer Science & Engineering, Chitkara University Institute of engineering & Technology, Punjab, India.

akumar@chitkara.edu.in

Abstract:

The rapid expansion of smart grids, renewable energy systems, and distributed power generation is reshaping the energy sector and opening the door to more decentralized ways to manage energy. Even so, most energy distribution still depends on centralized grid architectures. These systems can be inefficient, incur high transmission losses, offer limited transparency, and remain dependent on intermediaries. As a result, settlements may be delayed, consumers often have little control, and the overall infrastructure can be exposed to cyberattacks or single points of failure. Tools like smart meters and cloud-based monitoring platforms have improved automation and visibility, but they still place authority in centralized entities. This research proposes a decentralized peer-to-peer (P2P) energy trading framework using Ethereum blockchain technology. The system relies on smart contracts written in Solidity to automatically execute energy transactions between producers and consumers without intermediaries. MetaMask is considered for secure authentication and transaction authorization, while Web3.js can be used to connect a user interface to the blockchain network. Each transaction would be recorded immutably on the blockchain, supporting transparency, security, and trust for all participants. The study examines the feasibility, benefits, and potential impact of blockchain-based energy trading compared with traditional

centralized approaches. It also highlights how decentralization can strengthen transparency, lower intermediary costs, improve data integrity, and help enable future smart grid ecosystems.

Keywords:

Blockchain, Peer-to-Peer Energy Trading, Smart Contracts, Ethereum, Decentralized Energy Systems.

1. INTRODUCTION

With the rapid development of the energy sector, among which conventional and renewable energy sources are involved, there is a good chance that they will both play important roles in improving energy management to meet growing global energetic demand. Conventional energy systems are centralized where electricity is produced on a large scale by power plants and later these generated powers transferred to so many end users via top-down grid. Although this model has served well for decades, it faces several disadvantages like transmission losses, lack of transparency, a high operational cost and centralized authority dependence [21], [22].

In the traditional utility model, consumers are passive users of the system; however with smart grids and distributed energy resources like solar panels, wind turbines, etc., they have become active producers of electrical energy as well (these electricity generation-driven consumers are termed "prosumers"). These prosumers produce excess energy they can share or sell to others.

Despite wide strides achieved in smart grid-governed ecosystems and the increasing adoption of renewable technologies, current energy systems find it difficult to accommodate the decentralized energy trading that such changes have brought. This limitation mainly occurs due to their dependence on the conventional centralized control mechanisms, which results in inefficiencies, low transparency and trust among the participants. In addition, secure and peer-to-peer transaction models that enable surplus energy to be seamlessly exchanged between prosumers and consumers are lacking, underscoring the need for a more distributed solution that eliminates intermediaries [16].

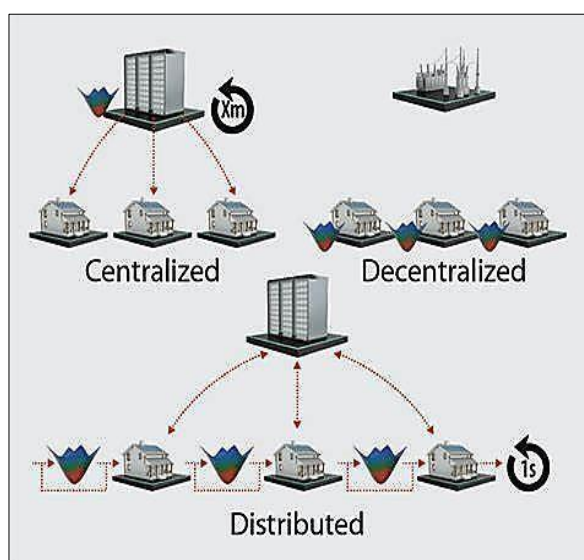


Figure 1.1: Comparison of Centralized, Decentralized, and Distributed Systems

A revolutionary technology called blockchain has been designed to solve these problems. [(Witko et al., 2018) One is a distributed ledger technology designed to store transactions securely, transparently, and without an intermediary. Implementing Blockchain with the energy systems provides scope for developing decentralized peer-to-peer(P2P) energy trading platforms where users can trade with each other directly.

The study aims at the development of a blockchain-based energy trading system that applies Ethereum smart contracts for automated transactions between energy producers and consumers [2],[3]. This

allows for increased transparency, efficiency and the removal of intermediaries in energy distribution. The proposed system data is based on decentralised technologies that aid in building sustainable and intelligent scenarios of energy ecosystems.

2. LITERATURE REVIEW

A number of studies have discussed the use of blockchain technology in the energy sector especially in facilitating decentralized and secure systems of energy trading. The initial research in blockchain proposed the notion of the distributed registry that provides transparency, immutability, and trustless transactions, which underlies and supports the development of multiple decentralized applications beyond digital currency [1].

Further studies have been made to utilize blockchain in terms of peer-to-peer (P2P) energy trading in microgrids. Previous research has shown that decentralized platforms may enable direct energy buying and selling between producers and consumers without the use of intermediaries, enhancing transparency and lowering the cost of running and making transactions [4], [6].

Moreover, frameworks in the form of smart contracts have been offered to manage energy transactions in an automated way. These schemes apply blockchain systems to implement predetermined terms to make sure that the transactions between the participants are safe, reliable, and impossible to tamper with, and that centralized control instruments are not required [3], [9].

In addition, detailed reviews of the blockchain use in the energy sphere suggest that it has a high possibility of improving the efficiency of the system, the safety of data, and sustainability. These articles emphasize how blockchain can radically change the traditional energy systems and how it can facilitate decentralized energy markets [5].

Nevertheless, even with these developments, there are a number of challenges. The current models are usually conceptual and small-scale but pay little attention to scalability, interoperability, and practical implementation in the real world. Critical concerns like elevated transaction expenses, network delays and regulatory bottlenecks are often under-explored [10].

Thus, it is apparent that a more viable and efficient decentralized energy trading solution is required that can overcome these shortcomings. The proposed study seeks to fill these gaps by suggesting a better model, which is more scalable, has less transaction overhead and is more real-life applicable.

3. OBJECTIVES

The main aim of this study is to design and come up with a decentralized peer-to-peer (P2P) energy trading system based on blockchain technology. The suggested system will reduce the reliance on centralized authorities and intermediaries, which will enhance efficiency and reliability of energy transactions.

This paper also dwells on the application of smart contracts on Ethereum blockchain to automate energy trading between producers and consumers. In this way, the system guarantees safe, transparent, and non-tampered transactions with no human involvement.

One more goal that is fundamental in nature is to improve transparency and trust amongst the participants and track all transactions on a distributed immutable ledger. Another goal of this research is to decrease the operational costs and transaction delays experienced in traditional energy systems.

Furthermore, the research aims to facilitate efficient consumption of renewable energy resources by allowing users to sell excess energy in a distributed setting. The main goal is to assess the viability and effectiveness of blockchain

technology in transforming traditional energy systems into smart, sustainable, and user-driven ecosystems.

4. METHODOLOGY

4.1. System Architecture

Here the proposed system is a decentralized peer-to-peer energy trading system based on blockchain [1]. The architecture contains several elements such as energy producers, energy consumers, web-based user interface, blockchain infrastructure and smart contracts.

Producers (prosumers) add excess energy to be sold on the platform and consumers can search for posted items and make a purchase from producers. A web-based front end communicates with the Smart Contract via Web3 in Ethereum.js [2], [19]. MetaMask functions as a digital wallet [18] to authenticate users and facilitate secure transactions.

The core logic of the system are smart contracts deployed on top of Ethereum blockchain [2], [3]. These contracts manage energy listing, transaction validation, and payment execution without the need for third parties. Transactions conducted within the network are recorded on a blockchain ledger, providing transparency, immutability and trust among participants.

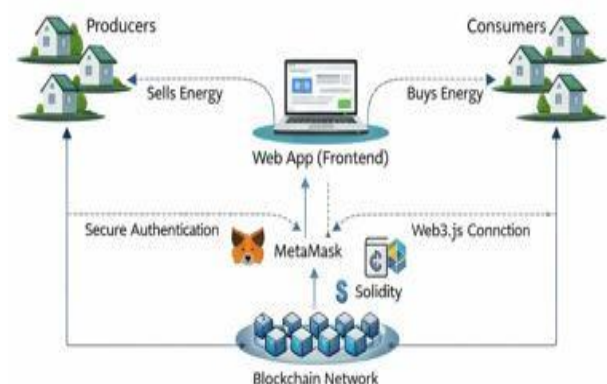


Figure 4.1: Blockchain-based Energy Trading System Architecture

4.2. Smart Contract Design

Solidity [17] is used to implement smart contracts, which outline the rules governing the energy trading process. The contract also has functions for listing energy, purchasing energy and validating transactions.

When an energy producer signs and lists a token of energy they are generating, the specifications (amount, pricing etc.) of that item will be saved on a blockchain." The contract allows consumers to interact and purchase energy by sending the required amount of cryptocurrency. The contract verifies the transaction conditions, such as sufficient balance and availability of energy, before executing the trade.

This automated execution eliminates the need for third-party intervention and ensures secure and tamper-proof transactions.

4.3. Blockchain Integration

The system integrates blockchain functionality using Web3.js, which acts as a bridge between the frontend application and the Ethereum network. Web3.js enables the application to interact with deployed smart contracts, retrieve data, and execute transactions in real time.

MetaMask is used for user authentication and transaction authorization. It allows users to securely manage their blockchain accounts and digitally sign transactions. Once a transaction is initiated, it is broadcast to the Ethereum network, where it is validated by nodes and permanently recorded on the blockchain.

4.4. System Workflow

The workflow of the proposed system follows a structured sequence of steps:

1. The user logs into the system using MetaMask authentication.
2. Producers list surplus energy along with pricing details.
3. Consumers browse available energy listings on the platform.

4. A consumer selects a listing and initiates a transaction.
5. The smart contract validates the transaction conditions.
6. Upon successful validation, the smart contract autonomously transfers the agreed digital assets (tokens/cryptocurrency) and updates ownership of the energy unit between the producer and consumer.
7. The transaction is immutably recorded on the blockchain ledger, including details such as participant addresses, transaction value, timestamp, and energy units exchanged. Both parties receive confirmation of the transaction.

This workflow ensures a seamless and transparent trading process, enabling direct interaction between producers and consumers.



Figure 4.2: Energy Trading workflow

4.5. Data Flow and Storage

The system follows a hybrid data storage approach. Transaction-related data, such as energy trades and payment records, are stored on the blockchain to ensure immutability and transparency [1]. Off-chain storage is used for managing user-related information and application data to improve system efficiency and scalability.

The data flow involves interactions between users, the application interface, smart contracts, and the blockchain network. Each transaction is validated through decentralized consensus mechanisms before being permanently recorded.

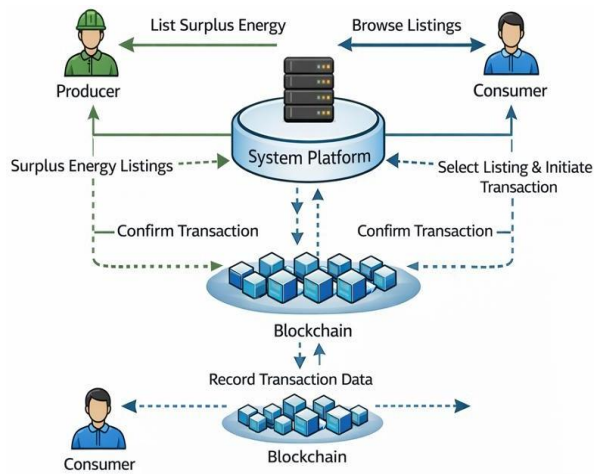


Figure 4.3: Data Flow Diagram of the System

4.6. Security and Reliability

Security is a key aspect of the proposed system. Blockchain technology ensures data integrity through cryptographic hashing and decentralized validation. Each transaction is verified by multiple nodes, reducing the risk of fraud and tampering.

Additionally, the use of smart contracts minimizes human intervention, thereby reducing errors and increasing system reliability. The decentralized nature of the system eliminates single points of failure [11], making it more robust compared to traditional centralized systems.

5. RESULTS

The blockchain-based peer-to-peer (P2P) energy trading system that we propose in this paper achieves obvious advantages when compared with conventional centralized energy delivery models in terms of efficiency, cost reduction, transparency and security [5], [8]. From results of conceptual analysis and system design evaluation, Key Performance Indicators (KPIs) were identified for assessing effectiveness with the proposed approach.

One of the major advantages observed is the reduction in intermediary costs. Traditional energy systems involve multiple intermediaries such as utility providers and billing agencies, which increase operational

expenses. By eliminating these intermediaries through smart contracts, the proposed system is estimated to reduce overall transaction and operational costs by approximately 20–30%, depending on network scale and transaction volume.

It allows these energy transactions to run without human interference through smart contracts. In contrast to traditional systems which might require hours or even days for a transaction to be settled, transactions on the blockchain can occur in as little as few seconds to few minutes (depending on the congestion level of the network). This results in an estimated 40–60% improvement in transaction processing time, enabling near real-time energy exchange.

Another important aspect is the gas fee associated with Ethereum transactions. While blockchain eliminates intermediary costs, it introduces transaction fees (gas fees) required for executing smart contracts. These fees can vary based on network demand and may become high during peak usage. This creates a trade-off between decentralization and cost efficiency. However, the impact of gas fees can be mitigated by using optimized smart contract design, Layer-2 scaling solutions, or alternative low-cost blockchain networks.

The system also shows strong performance in terms of transparency and trust. All transactions are recorded on a distributed ledger, allowing participants to verify and audit transactions in real time. This significantly reduces the risk of fraud and increases accountability within the system.

From a security perspective, the decentralized architecture ensures robustness against cyberattacks and system failures. The absence of a single point of failure enhances system reliability, making it more resilient compared to centralized infrastructures [14].

Additionally, the model promotes efficient utilization of renewable energy resources by enabling prosumers to trade surplus energy

directly. This not only reduces energy wastage but also contributes to a more sustainable and eco-friendly energy ecosystem.

However, the evaluation also highlights certain limitations. Scalability challenges may arise as the number of users and transactions increases, potentially affecting system performance. Furthermore, high gas fees and network latency in public blockchains like Ethereum may hinder large-scale adoption without further optimization.

Overall, the results indicate that the proposed system provides a cost-efficient, faster, and more transparent alternative to traditional energy trading models, with strong potential for real-world implementation after addressing scalability and cost-related challenges.

6. CONCLUSION

This research presents a blockchain-based decentralized peer-to-peer (P2P) energy trading system designed to overcome the limitations of traditional centralized energy distribution models. By leveraging Ethereum blockchain technology and smart contracts, the proposed system enables secure, transparent, and automated energy transactions between producers and consumers without the need for intermediaries.

The study shows that the use of blockchain technology can improve transparency, data integrity and trust among participants [5]. Smart Contracts are being used to access, monitor and verify between two parties counts the transactions, accuracy eliminating delays, reducing human error & alteration. This makes the system more resilient by removing single points of failure and allowing for a change to multiple attacks.

The proposed model also promotes the efficient utilization of renewable energy

resources by allowing users to trade surplus energy in a flexible and decentralized environment. This contributes to the development of sustainable and user-driven energy ecosystems aligned with future smart grid infrastructures.

Although certain challenges such as scalability and transaction costs remain, the overall findings confirm that blockchain technology has strong potential to revolutionize energy trading systems. With further advancements and real-world implementation, the proposed approach can play a significant role in shaping the future of decentralized and intelligent energy management [25].

Table 6-1: Comparison of Traditional vs Blockchain-Based Energy Trading

Traditional System	Energy	Blockchain-Based System
Centralized		Decentralized(P2P)
Multiple brokers)	(utilities,	None (smart contracts)
High		Reduced (~20–30%)
Slow(hours to days)		Fast (seconds to minutes)
Vulnerable to single point failure		Highly secure (distributed network)
Inefficient		Efficient (supports surplus trading)
Fixed operational costs		Variable (gas fees)

7. FUTURE SCOPE

Although the proposed blockchain-based energy trading system shows notable potential regarding secure and decentralized peer-to-peer energy exchange, there are multiple prospects which remain a topic for further research and development.

Firstly, scalability enhancement is a critical direction for future work. As the number of users and transactions increases, blockchain networks may experience latency and congestion. Future research can explore advanced solutions such as Layer-2 scaling techniques, sharding [10], or alternative

high-performance blockchain platforms to improve system efficiency.

Secondly, integration with smart grids and IoT devices presents a promising opportunity. Incorporating smart meters and IoT-enabled sensors can enable real-time energy monitoring, automated data collection, and dynamic pricing mechanisms, thereby improving the accuracy and responsiveness of the trading system.

Another important area is cost optimization. High transaction (gas) fees on blockchain platforms can limit adoption. Future work can investigate cost-effective consensus mechanisms or hybrid architectures that combine on-chain and off-chain processing to reduce operational expenses [8].

Additionally, regulatory and legal compliance remains a significant challenge. Future research should focus on aligning decentralized energy trading models with government policies and energy regulations to ensure practical deployment in real-world scenarios [22], [25].

Furthermore, interoperability between different blockchain networks can be explored to enable seamless communication and data exchange across multiple platforms, enhancing system flexibility and adoption.

Finally, the incorporation of advanced technologies such as artificial intelligence and machine learning can improve demand prediction, optimize energy distribution, and support intelligent decision-making within the system.

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